

# Spaceship Titanic Kaggle Competition Proposal

Machine Learning Classification Project

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## 1. Introduction

The Spaceship Titanic competition is a machine learning classification challenge hosted on Kaggle. The competition is inspired by a futuristic science-fiction scenario where an interstellar passenger ship collided with a spacetime anomaly, causing several passengers to be transported into another dimension.

The objective of this project is to build an intelligent predictive machine learning system capable of identifying whether a passenger was transported during the anomaly based on available passenger records. The dataset contains demographic information, spending behavior, cabin information, cryogenic sleep status, and destination details of passengers.

This competition provides an excellent opportunity to learn and apply real-world data science concepts including data preprocessing, exploratory data analysis, feature engineering, classification algorithms, ensemble learning, and model optimization.

## 2. Problem Statement

The Spaceship Titanic experienced a catastrophic collision with a spacetime anomaly during its journey to habitable exoplanets. Due to this event, many passengers were transported to an alternate dimension.

The goal of this project is to predict the binary target variable:

$$Transported = True \quad or \quad False$$

using the passenger information provided in the dataset.

The prediction model should accurately classify passengers who were transported and those who remained unaffected.

## 3. Objectives

The primary objectives of this project are:

- To understand and analyze the Spaceship Titanic dataset.
- To perform data preprocessing and handle missing values.
- To conduct exploratory data analysis (EDA).
- To engineer meaningful features from raw data.
- To build and compare multiple machine learning models.
- To improve prediction accuracy using ensemble techniques.
- To generate a valid Kaggle competition submission.

## 4. Dataset Description

The dataset contains passenger records recovered from the damaged spaceship computer system.

## 4.1 Training Dataset

The training dataset contains approximately 8700 passenger records along with the target variable.

## 4.2 Testing Dataset

The testing dataset contains approximately 4300 passenger records without target labels. Predictions must be generated for these passengers.

## 5. Features of the Dataset

| Feature      | Description                                       |
|--------------|---------------------------------------------------|
| PassengerId  | Unique passenger identifier in the format gggg-pp |
| HomePlanet   | Passenger's home planet                           |
| CryoSleep    | Whether passenger was in suspended animation      |
| Cabin        | Cabin identifier in the format deck/num/side      |
| Destination  | Destination planet                                |
| Age          | Age of the passenger                              |
| VIP          | VIP service status                                |
| RoomService  | Spending on room service                          |
| FoodCourt    | Spending on food court                            |
| ShoppingMall | Spending on shopping mall                         |
| Spa          | Spending on spa services                          |
| VRDeck       | Spending on VR deck services                      |
| Name         | Passenger full name                               |
| Transported  | Target variable indicating transportation status  |

## 6. Methodology

The methodology for this project is divided into multiple phases.

### 6.1 Phase 1: Data Collection

The dataset will be downloaded from Kaggle and loaded into the Python environment using Pandas.

### 6.2 Phase 2: Data Preprocessing

The preprocessing phase includes:

- Handling missing values
- Removing inconsistencies
- Converting categorical variables
- Data cleaning

## 6.3 Phase 3: Exploratory Data Analysis

EDA will be performed to understand patterns and relationships in the data.

Important analyses include:

- Missing value analysis
- Target distribution
- Correlation analysis
- Spending behavior analysis
- Passenger group analysis

## 6.4 Phase 4: Feature Engineering

Several new features will be created to improve model performance.

### 6.4.1 Total Spending Feature

$$TotalSpend = RoomService + FoodCourt + ShoppingMall + Spa + VRDeck$$

### 6.4.2 Cabin Features

Cabin information will be split into:

- Deck
- Cabin Number
- Side

### 6.4.3 Group Features

Passenger groups will be extracted from PassengerId.

### 6.4.4 Name Features

Family names and surname-based features will be extracted.

## 6.5 Phase 5: Model Building

Different machine learning models will be trained and evaluated.

### 6.5.1 Models to be Used

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost
- LightGBM
- CatBoost

## 6.6 Phase 6: Cross Validation

Stratified K-Fold cross validation will be used to avoid overfitting and ensure robust evaluation.

## 6.7 Phase 7: Hyperparameter Tuning

Hyperparameter tuning techniques such as Grid Search and Random Search will be applied to improve model accuracy.

## 6.8 Phase 8: Ensemble Learning

Multiple models will be combined using ensemble methods such as:

- Voting
- Stacking
- Blending

## 7. Tools and Technologies

| Tool         | Purpose                   |
|--------------|---------------------------|
| Python       | Programming Language      |
| Pandas       | Data Manipulation         |
| NumPy        | Numerical Computation     |
| Matplotlib   | Data Visualization        |
| Seaborn      | Statistical Visualization |
| Scikit-learn | Machine Learning          |
| XGBoost      | Gradient Boosting         |
| LightGBM     | Boosting Framework        |
| CatBoost     | Categorical Boosting      |
| Kaggle       | Competition Platform      |

## 8. Evaluation Metric

The competition uses Classification Accuracy as the evaluation metric.

$$Accuracy = \frac{CorrectPredictions}{TotalPredictions}$$

The objective is to maximize prediction accuracy on unseen test data.

## 9. Project Roadmap

| Week   | Task                                    |
|--------|-----------------------------------------|
| Week 1 | Dataset understanding and preprocessing |
| Week 2 | Exploratory Data Analysis (EDA)         |
| Week 3 | Feature engineering and transformation  |
| Week 4 | Baseline machine learning models        |
| Week 5 | Hyperparameter tuning                   |
| Week 6 | Ensemble learning and optimization      |
| Week 7 | Error analysis and final improvements   |
| Week 8 | Final submission and documentation      |

## 10. Expected Outcomes

The expected outcomes of this project include:

- Development of a high-accuracy classification model.
- Better understanding of feature engineering techniques.
- Practical experience with Kaggle competitions.
- Improved knowledge of machine learning workflows.
- Real-world problem solving using data science.

## 11. Future Scope

Future improvements may include:

- Deep learning approaches
- Advanced ensemble models
- Automated machine learning (AutoML)
- Feature selection optimization
- Explainable AI techniques

## 12. Conclusion

The Spaceship Titanic competition is an excellent platform for learning practical machine learning and data science concepts. Through this project, various stages of the machine learning lifecycle including preprocessing, feature engineering, model building, validation, and optimization will be explored.

The project aims to develop an efficient predictive model capable of accurately classifying transported passengers while also enhancing technical and analytical skills in machine learning.

## 13. References

1. Kaggle Spaceship Titanic Competition: <https://www.kaggle.com/competitions/spaceship-titanic>
2. Scikit-learn Documentation: <https://scikit-learn.org>
3. XGBoost Documentation: <https://xgboost.readthedocs.io>
4. LightGBM Documentation: <https://lightgbm.readthedocs.io>
5. CatBoost Documentation: <https://catboost.ai>